

Two Faces of Data:

Productivity and Markups in the Digital Age

Isaac Baley¹, Alejandro Rábano² and Julia M. Schmidt³

¹UPF Barcelona ²Banco de España ³Université Paris Dauphine–PSL

19 September 2026

International Workshop: AI in the Product, Factor and Financial Markets
Urbino, Italy

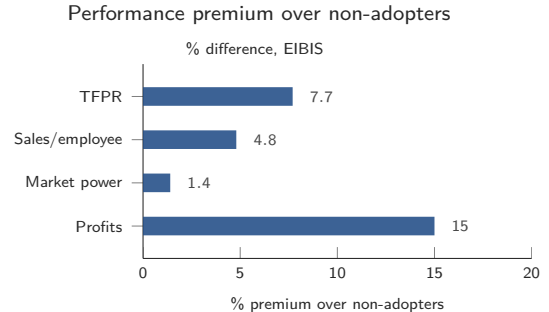
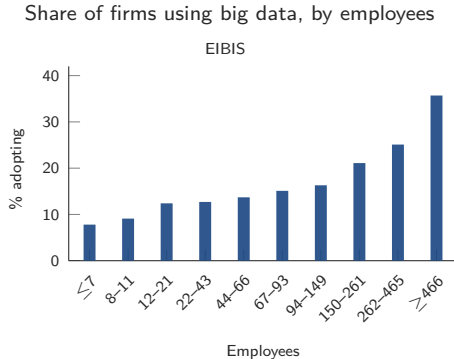
Background

Firm inequality has risen

- Between-firm dispersion has widened on *two* margins:
 - **Productivity**: frontier firms pulled ahead; digital capital concentrated in “superstar” firms (Andrews, Criscuolo, and Gal 2016; Tambe and Hitt 2021)
 - **Markups & concentration**: risen, driven by the top (De Loecker, Eeckhout, and Unger 2020); IT-driven in France (Lashkari, Bauer, and Boussard 2024)
- Standard view: the productive firms *are* the powerful firms, both rise together
- Is there a source of **markup dispersion** *distinct from* **productivity**?

Big data is concentrated among large firms and linked to performance

Big data adoption: use of technologies that automate tasks and provide insights, e.g. machine learning, robotic process automation, natural language processing, and neural networks.



EIB Investment Survey · Rábano, Rückert & Weiss (2025)

Data in this paper

Data is digitised information that reduces uncertainty (Baley and Veldkamp 2025); it

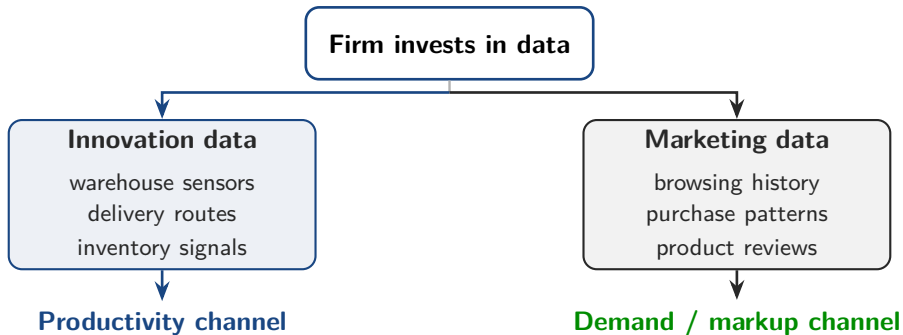
- improves performance by aiding decision-making
- loses relevance over time (depreciation)
- requires specialised labour to be utilised

We model two distinct types of data:

Ω **Innovation data**: firms learn how to produce efficiently

Ψ **Marketing data**: firms learn consumer preferences

Example: One Firm, Two Data Uses



The same firm can use data for two distinct purposes, with different implications for productivity and market power.

This Paper

Research question

Does data widen firm inequality through productivity, through market power, or both, and does it matter which?

What we do

- **Theoretical Model:** economy with heterogeneous firms that invest in two types of data (innovation and marketing data)
- **Evidence:** French TIC enterprise survey linked to firm-level balance sheets (FARE); firms report data use for production vs. consumer profiling; we estimate markups and TFP

Key Findings

Model

- **Markups vs. productivity:** the most **productive** firms are not automatically the high-**markup** firms
- **Substitution:** better **innovation technology** → fewer innovation workers
- **Classic patterns hold:** productive firms price lower, produce and earn more

Data (French firms, correlations)

- **Production data** ↔ higher **TFPR**; **consumer data** ↔ higher **markups**

Related Literature

Related Literature

Data and firms (Asriyan and Kohlhas 2025; Eeckhout and L. Veldkamp 2022; Farboodi and L. Veldkamp 2021): data is a *single* information technology, so data-rich (large) firms become *both* more productive and more powerful (via price discrimination, risk).

We model two *distinct* data inputs, with market power from a genuinely lower demand elasticity (real differentiation, not price discrimination). Data thus drives markup dispersion *distinct from* productivity.

- **Advertising and innovation** (Cavenaile and Roldan-Blanco 2021; Cavenaile, Celik, et al. 2025; Pearce and Wu 2025): R&D vs. advertising (persuasion, brand capital), consumer capital uncorrelated to productivity. Data as input in both types of *learning*; intra-firm allocation
- **Firm heterogeneity and market power** (Atkeson and Burstein 2008; Pellegrino 2025; Bornstein and Peter 2025): from product specialisation or non-linear pricing, information absent. **We** show *data* can be a source markup dispersion.

Model

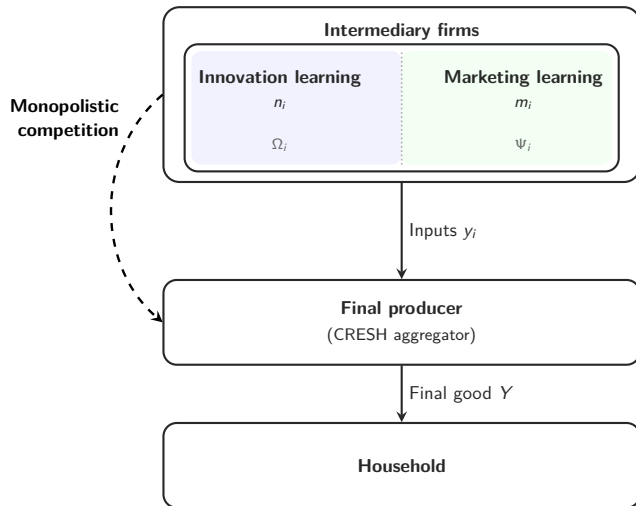
Model Intuition

- Firms compete in a monopolistic market for differentiated goods
- Every firm i allocates workers across **three activities**:
 - **Production** ℓ_i : make the good today
 - **Innovation** n_i : learn how to produce more efficiently $\rightarrow$ higher productivity A_i
 - **Marketing** m_i : learn what consumers want $\rightarrow$ lower demand elasticity $\rightarrow$ higher markup μ_i
- Firms differ in their **ability to learn** on each margin (z_i^A, z_i^M)
 $\rightarrow$ persistent heterogeneity in productivity and markups
- Three labour markets clear in **general equilibrium** $\rightarrow$ firm-level data decisions aggregate into wages and labour composition

Model Environment

► Household

► Market Clearing



- Each firm i faces a downward-sloping demand from the final producer:

$$y_i = \left(\frac{p_i}{P} \Gamma \frac{\bar{\gamma}_i}{\bar{\gamma}_i - 1} \right)^{-\bar{\gamma}_i} Y$$

Γ = demand index, taken as given by the firm.

- Demand elasticity $\bar{\gamma}_i$ is **firm-specific** and **endogenous**, not a common constant
- More marketing data $\psi_i \rightarrow$ lower $\bar{\gamma}_i \rightarrow$ higher markup $\mu_i = \frac{\bar{\gamma}_i}{\bar{\gamma}_i - 1} = \frac{1}{\bar{\lambda}_i}$

Information Accumulation

► Technology

► Signals

Firms learn by doing: specialist workers generate noisy signals that accumulate into data stocks. Signal precision τ_s is common, learning ability z_i is firm-specific, and existing data depreciates at rate ζ .

- **Innovation data**, from innovation workers n_i :

$$\Omega'_i = (1 - \zeta_A) \Omega_i + \tau_s^A z_i^A n_i^{\nu_A}$$

more $\Omega_i \Rightarrow$ less uncertainty about productivity $\Rightarrow \bar{A}_i \uparrow$

- **Marketing data**, from marketing workers m_i :

$$\Psi'_i = (1 - \zeta_M) \Psi_i + \tau_s^M z_i^M m_i^{\nu_M}$$

more $\Psi_i \Rightarrow$ less uncertainty about demand $\Rightarrow \bar{\gamma}_i \downarrow \Rightarrow$ markup $\uparrow$

Firm Problem

► Steady state

Choose production ℓ_i , innovation n_i , marketing m_i to maximise profit today plus discounted future value:

$$V_i(\Omega_i, \Psi_i; z_i^A, z_i^M) = \max_{\ell_i, n_i, m_i \geq 0} \left\{ \bar{R}_i - w_\ell \ell_i - w_n n_i - w_m m_i + \beta V_i(\Omega'_i, \Psi'_i; z_i^A, z_i^M) \right\}$$

$$\text{s.t. } \bar{y}_i = \bar{A}_i \ell_i, \quad \bar{R}_i = p_i \bar{y}_i$$

$$\Omega'_i = (1 - \zeta_A) \Omega_i + \tau_s^A z_i^A n_i^{\nu_A}, \quad \Psi'_i = (1 - \zeta_M) \Psi_i + \tau_s^M z_i^M m_i^{\nu_M}$$

Static optimum: the two channels meet in the price and thus in revenue:

$$p_i = \mu_i \frac{w_\ell}{\bar{A}_i} \quad (\text{productivity sets cost, markup is the wedge})$$

What Drives Data Investment?

► Innovation FOC

► Marketing FOC

Innovation n_i rises when:

- Revenue $\bar{R}_i$ is large: more output to apply higher TFP to
- Data Ω_i is scarce: each signal matters more

Marketing m_i rises when:

- Data Ψ_i is scarce: each signal matters more
- **Room to differentiate**: the firm is small relative to the market

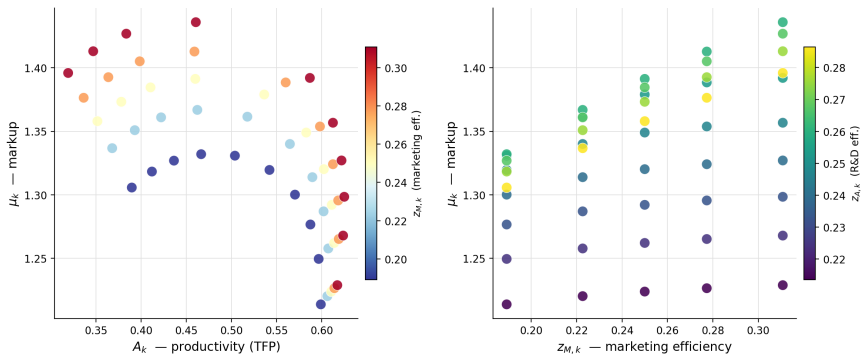
The asymmetry: scale *drives* innovation but *erodes* marketing, a big, productive firm invests heavily in innovation yet little in marketing.

Results

Productive Firms Are Not Automatically High-Markup Firms

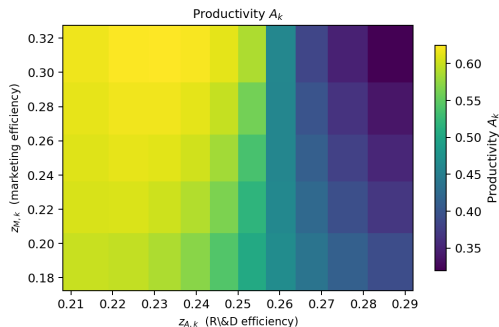
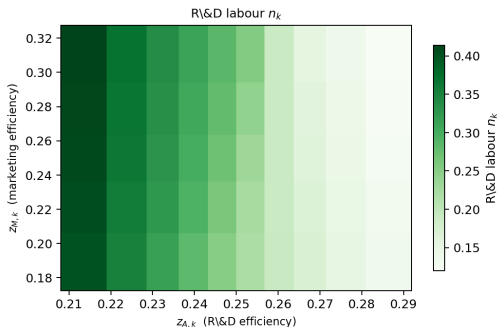
► Mechanism

Firms differ in two data technologies: innovation z_i^A and marketing z_i^M .



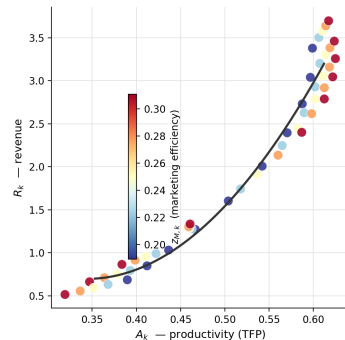
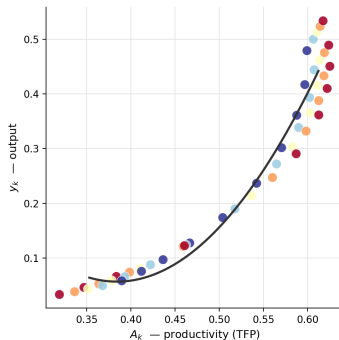
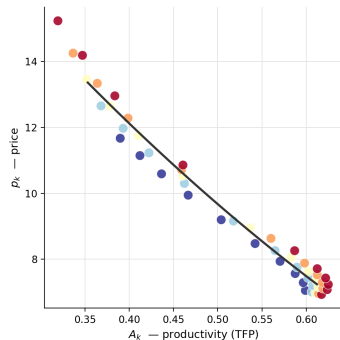
At a given productivity, markups are ordered by marketing technology z_i^M (colour); a productive firm has a high markup only if it is also good at learning demand.

Substitution: Better Innovation Technology, Lower Productivity



Higher z^A reaches the same knowledge with fewer workers, so firms hire less innovation labour (left) and end up with lower productivity (right).

Classic Patterns Hold: Price, Output, and Revenue



More productive firms price lower, produce and earn more, whatever their technology (colour).

Empirical Evidence

- **ICT Enterprise Survey** (INSEE), firms ≥ 10 employees, 2015, 2017, 2019: Big Data use by source
 - **Consumer data**: social media
 - **Production data**: sensors, smart devices, geolocation
- **FARE balance sheets** (INSEE), matched on SIREN: sales, employment, capital, costs
- $\sim 20,000$ firms, survey-weighted

Each Data Type Is Used for Its Own Purpose

Logit marginal effects: probability a firm uses data for each purpose (2015 Big Data adopters)

► Sectors

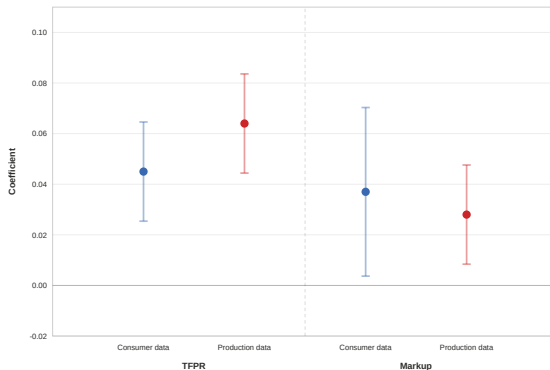
	<i>Data used for</i>		
	Marketing	Process	Product
Consumer data	0.294*** (0.05)	−0.037 (0.04)	0.236*** (0.05)
Production data	0.060 (0.05)	0.252*** (0.05)	0.250*** (0.04)
Observations	994	994	994
Sector FE, capital, age, other tech.	Yes	Yes	Yes

Consumer data → marketing, not process; production data → process, not marketing. Both feed product development.

Data Types, Productivity and Markups

$$y_{ist} = \alpha_{st} + \beta_0 \text{Data Type}_{ist} + \beta_1 k_{ist} + \beta_2 \text{age}_{ist} + \gamma' \Theta_{ist} + \epsilon_{ist}$$

y: TFPR (Akerberg, Caves, and Frazer 2015) or markup (Loecker and Warzynski 2012); separate regressions by data type; sector-year FE; within-sector correlations, not causal



Conclusion

Summary

- Data can widen firm inequality through **productivity** and **market power**, and these are *different* margins
- **Model:** two data uses, **innovation** and **marketing**
 - the most productive firms are not automatically the high-markup firms
 - better innovation technology can *lower* productivity (substitution)
 - classic firm-size patterns still hold
- **Data:** each data type is used for its own purpose; **production data** correlates more with **TFPR**, **consumer data** more with **markups**
- **Policy:** data diffusion and governance should depend on whether data creates **efficiency gains** or **market-power gains**

Next Steps

- **Calibration:** discipline (z^A, z^M) with French firm data
- **Aggregate uncertainty shock:** when uncertainty rises, do firms shift specialists between innovation and marketing?
- **Sharper empirics:**
 - test the *gaps* between coefficients jointly
 - markup robustness following (De Ridder, Grassi, and Morzenti 2026)
- **Counterfactual, GDPR:** restricting consumer data shifts the z^M distribution down.

Thank you!

Julia M. Schmidt

julia.schmidt@dauphine.psl.eu

References i

-  Akerberg, Daniel A., Kevin Caves, and Garth Frazer (2015). **“Identification Properties of Recent Production Function Estimators”**. en. In: *Econometrica* 83.6, pp. 2411–2451. ISSN: 0012-9682.
-  Andrews, Dan, Chiara Criscuolo, and Peter N. Gal (Dec. 2016). **“The Best versus the Rest: The Global Productivity Slowdown, Divergence across Firms and the Role of Public Policy”**. en. In: *OECD Productivity Working Papers*.
-  Asriyan, Vladimir and Alexandre Kohlhas (2025). **“The Macroeconomics of Data: Scale, Product Choice, and Pricing in the Information Age”**. en. In.
-  Atkeson, Andrew and Ariel Burstein (Nov. 2008). **“Pricing-to-Market, Trade Costs, and International Relative Prices”**. en. In: *American Economic Review* 98.5, pp. 1998–2031. ISSN: 0002-8282.

References ii

-  Baley and Veldkamp (Jan. 2025). ***The Data Economy*** / Princeton University Press. en. ISBN: 978-0-691-25672-6.
-  Bornstein, Gideon and Alessandra Peter (Nov. 2025). “**Nonlinear Pricing and Misallocation**”. en. In: *American Economic Review* 115.11, pp. 3853–3908. ISSN: 0002-8282.
-  Cavenaile, Laurent, Murat Alp Celik, et al. (Jan. 2025). “**Style over substance? Advertising, innovation, and endogenous market structure**”. en. In: *Journal of Monetary Economics* 149, p. 103683. ISSN: 03043932.
-  Cavenaile, Laurent and Pau Roldan-Blanco (July 2021). “**Advertising, Innovation, and Economic Growth**”. en. In: *American Economic Journal: Macroeconomics* 13.3, pp. 251–303. ISSN: 1945-7707, 1945-7715.

References iii

-  De Loecker, Jan, Jan Eeckhout, and Gabriel Unger (May 2020). “**The Rise of Market Power and the Macroeconomic Implications**”. In: *The Quarterly Journal of Economics* 135.2, pp. 561–644. ISSN: 0033-5533.
-  De Ridder, Maarten, Basile Grassi, and Giovanni Morzenti (2026). “**The Hitchhiker’s Guide to Markup Estimation: Assessing Estimates From Financial Data**”. en. In: *Econometrica* 94.1, pp. 137–168. ISSN: 0012-9682.
-  Eeckhout, Jan and Laura Veldkamp (2022). “**Data and Market Power**”. en. In.
-  Farboodi, Maryam and Laura Veldkamp (Feb. 2021). **A Model of the Data Economy**. en. Tech. rep. w28427. Cambridge, MA: National Bureau of Economic Research, w28427.

References iv

-  Lashkari, Danial, Arthur Bauer, and Jocelyn Boussard (June 2024). “**Information Technology and Returns to Scale**”. en. In: *American Economic Review* 114.6, pp. 1769–1815. ISSN: 0002-8282, 1944-7981.
-  Loecker, Jan De and Frederic Warzynski (Oct. 2012). “**Markups and Firm-Level Export Status**”. en. In: *American Economic Review* 102.6, pp. 2437–2471. ISSN: 0002-8282.
-  Pearce, Jeremy and Liangjie Wu (2025). “**Market Concentration and Productivity: The Role of Demand**”. en. In.
-  Pellegrino, Bruno (Apr. 2025). “**Product Differentiation and Oligopoly: A Network Approach**”. en. In: *American Economic Review* 115.4, pp. 1170–1225. ISSN: 0002-8282.
-  Tambe, Prasanna and Lorin M Hitt (2021). “**Digital Capital and Superstar Firms**”. en. In.

Appendix

$$\Gamma = \int_0^1 \lambda_i \left(\frac{y_i}{Y} \right)^{\lambda_i} di, \quad \lambda_i \equiv \frac{\gamma_i - 1}{\gamma_i}$$

Sector Heterogeneity in Use of Marketing and Production Data [▶ Back](#)

Sector	Consumer Data (% of firms)	Production Data (% of firms)
IT & Telecom	14.5	16.1
R&D and Adv	13.6	15.6
Tourism	7.1	4.1
Business Services	7.1	13.7
Consulting	6.3	12.9
Utilities	5.9	21.8
Food	5.5	6.0
Trade	5.2	7.4
Chemicals	4.9	8.2
Household Services	4.7	9.6
Electronics	4.7	12.5
Textiles	3.4	10.3
Construction	3.3	12.4
Transport	3.1	27.4
Materials	2.7	7.2
Machinery	2.6	9.2

Steady State (I) [▶ Back](#)

$$\text{Productivity: } \bar{A}_i = \mathcal{A}(\Omega_i)$$

$$\text{Demand elasticity: } \bar{\gamma}_i = \mathcal{G}(\Psi_i), \quad \bar{\lambda}_i \equiv 1 - \frac{1}{\bar{\gamma}_i}, \quad \mu_i = \frac{1}{\bar{\lambda}_i}$$

$$\text{Production: } \bar{y}_i = \bar{A}_i \ell_i$$

$$\text{Inverse demand: } p_i = P \frac{\bar{\lambda}_i}{\Gamma} \left(\frac{\bar{y}_i}{Y} \right)^{\bar{\lambda}_i - 1}, \quad \Gamma = \int_0^1 \bar{\lambda}_i \left(\frac{\bar{y}_i}{Y} \right)^{\bar{\lambda}_i} di$$

$$\text{Revenue: } \bar{R}_i = p_i \bar{y}_i = P \frac{\bar{\lambda}_i}{\Gamma} Y^{1-\bar{\lambda}_i} (\bar{A}_i \ell_i)^{\bar{\lambda}_i}$$

$$\text{Information law (innovation): } \zeta_A \Omega_i = \tau_s^A z_i^A n_i^{\nu_A}$$

$$P = 1 \text{ (numeraire). Gaussian learning: } \mathcal{A}(\Omega_i) = A_0 - [\Omega_i^{-1} + (\tau_\varepsilon^A)^{-1}], \quad \mathcal{G}(\Psi_i) = \gamma_0 + [\Psi_i^{-1} + (\tau_\varepsilon^M)^{-1}].$$

Steady State (II) [▶ Back](#)

Information law (marketing): $\zeta_M \Psi_i = \tau_s^M z_i^M m_i^{\nu_M}$

Production FOC (static): $p_i \bar{\lambda}_i \bar{A}_i = w_\ell$

Innovation FOC (dynamic): $w_n = \beta \tau_s^A z_i^A \nu_A n_i^{\nu_A-1} V_{\Omega,i}$

Marketing FOC (dynamic): $w_m = \beta \tau_s^M z_i^M \nu_M m_i^{\nu_M-1} V_{\Psi,i}$

Shadow values: $V_{\Omega,i} = \frac{\partial \bar{R}_i / \partial \Omega_i}{1 - \beta(1 - \zeta_A)}, \quad V_{\Psi,i} = \frac{\partial \bar{R}_i / \partial \Psi_i}{1 - \beta(1 - \zeta_M)}$

Marginal revenue of data: $\frac{\partial \bar{R}_i}{\partial \Omega_i} = \bar{R}_i \bar{\lambda}_i \frac{\mathcal{A}'(\Omega_i)}{\bar{A}_i}, \quad \frac{\partial \bar{R}_i}{\partial \Psi_i} = -\frac{\bar{R}_i \mathcal{G}'(\Psi_i)}{\bar{\gamma}_i^2} \left[\log\left(\frac{Y}{\bar{y}_i}\right) - \mu_i \right]$

$\bar{y}_i \equiv \bar{A}_i \ell_i, \quad \mu_i = 1/\bar{\lambda}_i$. Gaussian learning: $\mathcal{A}'(\Omega_i) = \Omega_i^{-2}, \quad \mathcal{G}'(\Psi_i) = -\Psi_i^{-2}$.

Learning Micro-foundation (I): Why Data Raises $\bar{A}$ and Lowers $\bar{\gamma}$

[▶ Back](#)

1. Performance depends on hitting a moving target.

Innovation: $A_{it} = A_0 - (a_{it} - a_t^*)^2$ closer technique $\rightarrow A \uparrow$

Marketing: $\gamma_{it} = \gamma_0 + (g_{it} - g_t^*)^2$ closer positioning $\rightarrow \gamma \downarrow \rightarrow$ markup $\uparrow$

2. The target has a learnable and an unlearnable part.

$$a_t^* = \theta_t^A + \varepsilon_t^A, \quad \theta_{t+1}^A = \rho_A \theta_t^A + \xi_{t+1}^A \quad (\text{same for } g_t^*)$$

$\xi \sim \mathcal{N}(0, (\tau_\theta)^{-1})$: drift that data can track; $\varepsilon \sim \mathcal{N}(0, (\tau_\varepsilon)^{-1})$: noise that no data removes.

3. The firm aims at its best guess, $a_{it} = \mathbb{E}[\theta_t^A \mid \mathcal{I}_{it}]$. The expected miss is posterior variance plus noise:

$$\bar{A}_i = A_0 - \left[\Omega_i^{-1} + (\tau_\varepsilon^A)^{-1} \right], \quad \bar{\gamma}_i = \gamma_0 + \left[\Psi_i^{-1} + (\tau_\varepsilon^M)^{-1} \right]$$

Ω_i, Ψ_i = precision of beliefs. More data helps at a falling rate, and the noise floor bounds $\bar{A}$ and $\bar{\gamma}$.

How precision builds up: Signals.

Learning Micro-foundation (II): How Specialists Build Precision

[▶ Back](#)

Shown for innovation; marketing is identical with Ψ , m and superscript M .

1. Specialists produce signals about next period's target:

$$s_{it}^A = \theta_{t+1}^A + \eta_{it}^A, \quad \text{precision of } \eta_{it}^A = \tau_s^A z_i^A n_{it}^{\nu_A}, \quad \nu_A \in (0, 1)$$

More innovation workers n or a better technology $z^A \rightarrow$ a sharper signal; $\nu_A < 1 \rightarrow$ diminishing returns to specialists.

2. Kalman updating: forget, then learn.

$$\text{Forget (target drifts): } \Omega_{i,t+1|t} = \left(\rho_A^2 \Omega_{it}^{-1} + (\tau_\theta^A)^{-1} \right)^{-1}$$

$$\text{Learn (signal arrives): } \Omega_{i,t+1} = \Omega_{i,t+1|t} + \tau_s^A z_i^A n_{it}^{\nu_A}$$

3. Reduced form used in the model.

$$\Omega_i' = (1 - \zeta_A) \Omega_i + \tau_s^A z_i^A n_i^{\nu_A}$$

The depreciation rate ζ_A stands in for the forgetting step. Steady state.

Production Labour

► Back

The firm equates the marginal revenue product of production labour to the wage:

$$p_i \bar{\lambda}_i \bar{A}_i = w_\ell$$

Equivalently, the price is a firm-specific markup over marginal cost:

$$p_i = \mu_i \cdot \frac{w_\ell}{\bar{A}_i}, \quad \mu_i = \frac{1}{\bar{\lambda}_i}$$

- $\bar{A}_i \uparrow$ (higher productivity) $\Rightarrow$ lower marginal cost, more labour demanded
- μ_i is set by marketing data Ψ_i : more $\Psi_i \Rightarrow$ lower $\bar{\gamma}_i$ and $\bar{\lambda}_i \Rightarrow$ higher markup

Innovation Labour

► Back

Innovation labour is a dynamic choice: it raises future production knowledge Ω'_i , not current output. The optimality condition is

$$w_n = \frac{\beta \tau_s^A z_i^A \nu_A n_i^{\nu_A-1}}{1 - \beta(1 - \zeta_A)} \cdot \frac{\bar{R}_i \bar{\lambda}_i \mathcal{A}'(\Omega_i)}{\bar{A}_i}$$

The value of production knowledge is higher when:

- $\bar{R}_i \uparrow$: more revenue to gain from higher productivity
- $\bar{\lambda}_i \uparrow$: demand is *more elastic*, so a productivity gain passes into revenue more strongly. It is worth more where market power is *weaker*
- $\Omega_i \downarrow$: information is still scarce, so each unit of learning matters more
- $\bar{A}_i \downarrow$: productivity is far below its ceiling

Gaussian learning: $\mathcal{A}'(\Omega_i) = \Omega_i^{-2}$.

Marketing Labour

[▶ Back](#)

Marketing labour is a dynamic choice: it raises future demand knowledge Ψ'_i , which lowers the demand elasticity and raises the markup. The optimality condition is

$$w_m = - \frac{\beta \tau_s^M z_i^M \nu_M m_i^{\nu_M-1}}{1 - \beta(1 - \zeta_M)} \cdot \frac{\bar{R}_i \mathcal{G}'(\Psi_i)}{\bar{\gamma}_i^2} \left[\log\left(\frac{Y}{\bar{y}_i}\right) - \mu_i \right]$$

The value of demand knowledge is higher when:

- $\bar{R}_i \uparrow$: a lower elasticity applies to a larger revenue base
- $\Psi_i \downarrow$: demand knowledge is still scarce
- $\bar{y}_i$ small relative to Y : more room to differentiate

Scale enters *twice*, with opposite signs: more revenue, less room. If $\log(Y/\bar{y}_i) < \mu_i$, the firm hires no marketing workers.

$\bar{y}_i = \bar{A}_i \ell_i$, $\mu_i = 1/\bar{\lambda}_i$. Gaussian learning: $\mathcal{G}'(\Psi_i) = -\Psi_i^{-2}$.

Household

[▶ Back](#)

A representative household consumes C and splits one unit of time across production L , innovation N and marketing M :

$$\begin{aligned} \max_{L, N, M} \quad & \log C - \frac{\chi_\ell}{1+\varphi} L^{1+\varphi} - \frac{\chi_n}{1+\varphi} N^{1+\varphi} - \frac{\chi_m}{1+\varphi} M^{1+\varphi} \\ \text{s.t.} \quad & L + N + M = 1, \quad PC = w_\ell L + w_n N + w_m M + \Pi \end{aligned}$$

FOCs ($1/Y$ on the budget, κ on time; $P = 1$, $C = Y$):

$$\chi_j J^\varphi = \frac{w_j}{Y} - \kappa \quad (j = \ell, n, m)$$

Wage wedges (eliminating κ):

$$\begin{aligned} \frac{w_\ell - w_n}{Y} &= \chi_\ell L^\varphi - \chi_n N^\varphi \\ \frac{w_\ell - w_m}{Y} &= \chi_\ell L^\varphi - \chi_m M^\varphi \end{aligned}$$

Occupations compete for one time endowment: more marketing labour bids up w_m and makes production and innovation workers scarcer. Π = firm profits, rebated lump-sum.

Market Clearing & Equilibrium

► Back

Goods market: $C = Y$, $P = 1$ (numeraire).

Labour markets (aggregating over firm types):

$$L = \int \ell_i dG(z^A, z^M), \quad N = \int n_i dG(z^A, z^M), \quad M = \int m_i dG(z^A, z^M)$$

Profits: $\Pi = \int \pi_i dG(z^A, z^M)$, $\pi_i = \bar{R}_i - w_\ell \ell_i - w_n n_i - w_m m_i$

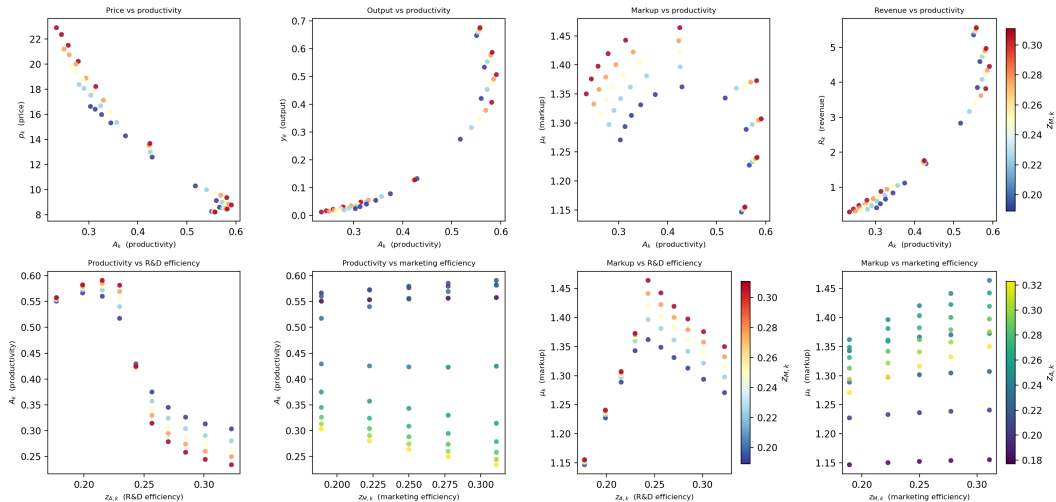
CRESH index: $\Gamma = \int_0^1 \bar{\lambda}_i (\bar{y}_i / Y)^{\bar{\lambda}_i} di$

Steady-state equilibrium: allocations $\{\ell_i, n_i, m_i\}$, wages (w_ℓ, w_n, w_m) and output Y :

1. the household maximises utility
2. the final producer minimises cost subject to the CRESH aggregator
3. every firm type maximises the value of profits, with (Ω_i, Ψ_i) at their steady-state levels
4. wages clear the three labour markets

Cross-sectional diagnostics in GE

Firm-level scatter: price / output / markup vs productivity and efficiency types (baseline het GE)



Mechanism: Two Ways of Being Large [▶ Back](#)

The marginal value of demand precision is a product of three terms:

$$\Lambda_i^M \propto \underbrace{\psi_i^{-2}}_{\text{satiation}} \times \underbrace{\frac{\bar{R}_i}{\bar{\gamma}_i^2}}_{\text{revenue base}} \times \underbrace{\left[\log(Y/\bar{y}_i) - \mu_i \right]}_{\text{room to differentiate}}$$

- Large *in quantity* (the productive firm): huge revenue base, but output sits close to $Y \Rightarrow$ no room to differentiate
- Large *in markup*: small in quantity, so room remains; the markup raises the revenue base faster than it erodes that room
- What caps power is *satiation*, not position: each further unit of precision buys a smaller elasticity cut